



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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LectureNotes-2.2

**22GE004–BASICS OF ELECTRONICS
ENGINEERING**

**UNIT II & SIGNAL CONDITIONING
USING DIODE**

22GE004 - BASICS OF ELECTRONICS ENGINEERING

UNIT II - SIGNAL CONDITIONING USING DIODE

1. Introduction:

Electronics have evolved significantly over the years, transitioning from vacuum tubes to transistors, and then to integrated circuits.

Vacuum tubes played a pivotal role in the development of early electronic devices, paving the way for modern electronics.

Objective:

To understand the significance of vacuum tubes in the evolution of electronics.

To provide an overview of vacuum tubes and their functionality.

i. Historical Context:

Brief history of vacuum tubes, dating back to the late 19th century.

Early experiments by inventors such as Thomas Edison and John Ambrose Fleming.

ii. Basics of Vacuum Tubes:

Definition of a vacuum tube: a glass tube containing electrodes from which air has been removed, allowing electrons to flow freely.

Components of a vacuum tube: cathode, anode (plate), and control grid.

How vacuum tubes work: emission of electrons from the heated cathode, control of electron flow by the grid, and amplification of signals.

iii. Applications of Vacuum Tubes:

Amplification: Early radios and televisions relied on vacuum tubes for signal amplification.

Switching: Vacuum tubes were used in early computers and industrial control systems for switching purposes.

Oscillation: Vacuum tubes were crucial components in the development of early oscillators and signal generators.

iv. Advantages and Disadvantages:

Advantages:

High power handling capabilities.

Relatively simple design.

Well-understood technology at the time.

Disadvantages:

Large size and bulky nature.

Fragility and limited lifespan.

High power consumption and heat generation.

v. Impact on Electronics Evolution:

Vacuum tubes laid the foundation for modern electronic devices.

They enabled the development of early electronic systems, including radios, televisions, and early computers.

Their limitations spurred the search for alternative technologies, leading to the invention of transistors.

vi. Transition to Transistors:

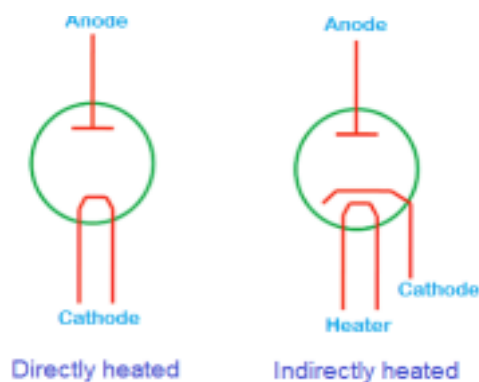
Invention of the transistor in 1947 by John Bardeen, Walter Brattain, and William Shockley.

Transistors offered several advantages over vacuum tubes, including smaller size, lower power consumption, and greater reliability.

The widespread adoption of transistors led to the decline of vacuum tube technology in most consumer electronics.

1.2 Vacuum diode:

Two Electrodes of Vacuum tubes:



Vacuum diode is the simplest form of vacuum tube. It consists of two electrodes, a cathode, and an anode or plate. The cathode emits the free electrons. Hence, it is called an emitter. The anode collects the free electrons. Hence, it is called a collector.

The cathode and anode are enclosed in an empty glass envelope. The anode is a hollow cylinder made of molybdenum or nickel and cathode is a nickel cylinder coated with strontium and barium oxide. The anode surrounds the cathode. In between the cathode and anode an empty space is present, through which the free electrons or electric current flow.

1.3 Electrode:

Electrode is a conductor through which free electrons or electric current leaves or enters. In vacuum diodes, cathode is an electrode or conductor from which the free electrons are emitted into the vacuum.

On the other hand, anode is an electrode that collects the free electrons emitted by the cathode. In other words, free electrons leave the cathode and enter into anode. Electron emission depends on the amount of heat applied and the work function.

The number of free electrons emitted by the cathode depends on two factors: amount of heat applied and work function.

If more heat is applied, a greater number of free electrons is emitted. Similarly, if less amount of heat is applied, a smaller number of free electrons is emitted. The minimum amount of energy required to remove the free electrons from the metal is called work function. Metals with low work function will require less amount of heat energy to emit the free electrons.

On the other hand, metals with high work function will require a large amount of energy to emit the free electrons. Hence, choosing a good material will increase the electron emission efficiency. Most commonly used thermionic emitters include oxide-coated cathode, tungsten, and thoriated tungsten.

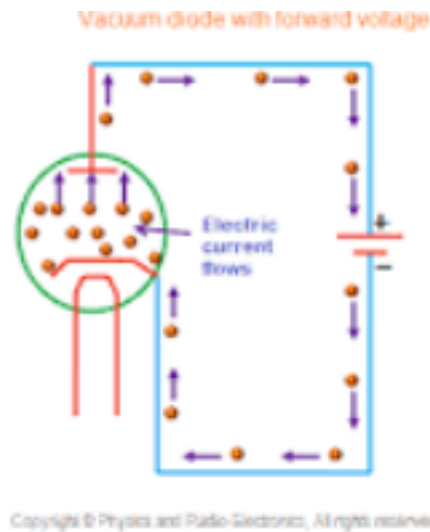
1.3.1 Directly and Indirectly heated Cathode:

When the cathode is indirectly or directly heated, free electrons are emitted from it.

- In the directly heated cathode, the heat energy is supplied directly to the cathode. Hence, a small amount of heat energy is enough to emit the free electrons from the cathode. When the heat energy is directly supplied to the cathode, a large number of free electrons gain sufficient energy and break the bonding with the cathode. The free electrons that break the bonding with the cathode are emitted into the vacuum. These emitted free electrons are attracted towards the anode.
- In the indirectly heated cathode, no electrical connection is present between the cathode and the heater. Hence, the cathode is not heated directly. The heat energy is supplied to the heater and the heater will transfer its heat energy to the cathode. When the heat energy applied to the cathode is increased to a desired level, the free electrons in the cathode gain sufficient energy and break the bonding with the cathode. The free electrons that break the bonding with the cathode are emitted into the vacuum. These emitted free electrons are attracted towards the anode.

1.3.2 Vacuum diode with forward voltage:

When the heat is supplied to the heater, it gains heat energy. This heat energy is transferred to the cathode. When the free electrons in the cathode gain sufficient energy, they break the bonding with the cathode and jump into vacuum. The free electrons in the vacuum need sufficient kinetic energy to reach the anode. If voltage is applied to the vacuum diode, in such a way, that anode is connected to a positive terminal and cathode is connected to a negative terminal (anode is more positive with respect to the cathode), the free electrons in the vacuum gain enough kinetic energy to reach the anode.



We know that, if two opposite charged particles are placed close to each other they get attracted. In this case, anode is positively charged and free electrons emitted from the cathode are negatively charged.

Hence, the free electrons that gain enough kinetic energy will move or attracted towards the anode. These free electrons carry the electric current while moving from cathode to anode.

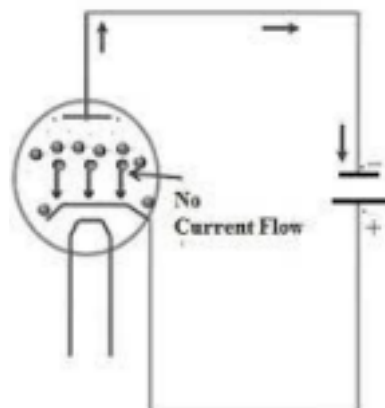
If the positive voltage applied to plate or anode is increased, the number of free electrons attracted towards the anode is also increased. Thus, the electric current in the vacuum diode increases with increase in the anode or plate voltage.

1.3.3 Vacuum diode with reverse voltage:

If voltage is applied to the vacuum diode, in such a way, that anode is connected to the negative terminal and cathode is connected to the positive terminal (anode is more negative with respect to cathode), the free electrons in the vacuum gains enough kinetic energy to reach the anode. However, anode repels the free electrons that try to move towards it.

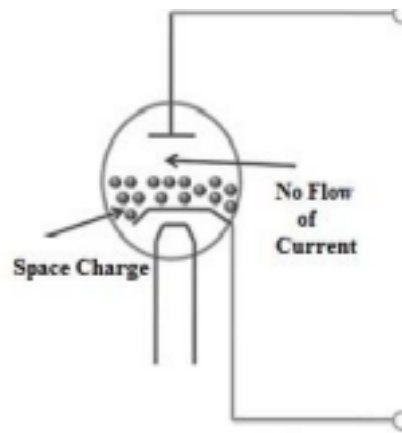
We know that if two like charged particles are placed close to each other they get repelled. In this case, anode is negatively charged and the free electrons emitted from the cathode are also negatively charged.

Hence, the anode repels the free electrons that are emitted by the cathode. Therefore, no electric current flows in the vacuum diode.



1.3.4 Vacuum diode with zero voltage:

If no voltage is applied to the vacuum diode, anode or plate acts as neutral. It neither attracts or repels the free electrons emitted from the cathode. Hence, the free electrons emitted from the cathode do not move or attract towards the anode.

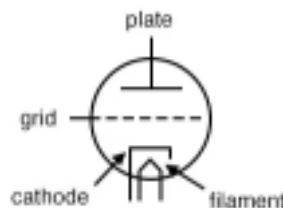


Therefore, no electric current occurs in the vacuum diode. However, the large number of free electrons emitted from the cathode builds up at one place near the cathode and forms a cloud of free electrons. This cloud of free electrons near the cathode is called space charge.

Therefore, the vacuum diode allows the electric current from cathode to anode and does not allow the electric current from anode to cathode. This one-way direction of the electric current enables the vacuum diode to act like a switch. If the anode or plate is positive with respect to the cathode, the vacuum diode acts like a closed switch. On the other hand, if the anode is negative with respect to the cathode, it acts like an open switch.

1.4 Vacuum triode:

The basic vacuum tube (vacuum diode) is used to convert the alternating current into direct current. However, they cannot amplify the electric signal. In other words, they cannot amplify the voltage or power. To amplify the electrical signal, an extra electrode is required. When the extra electrode is placed between the cathode and anode, the resulting electronic device is called vacuum triode. The name itself indicates that it has three electrodes: cathode, anode, and control grid. American electrical engineer Lee De Forest invented the first electronic amplifying device (vacuum triode) in 1906 by adding an extra electrode (control grid) between the cathode and anode. Vacuum triode is a 3-electrode device that amplifies the electrical signal.

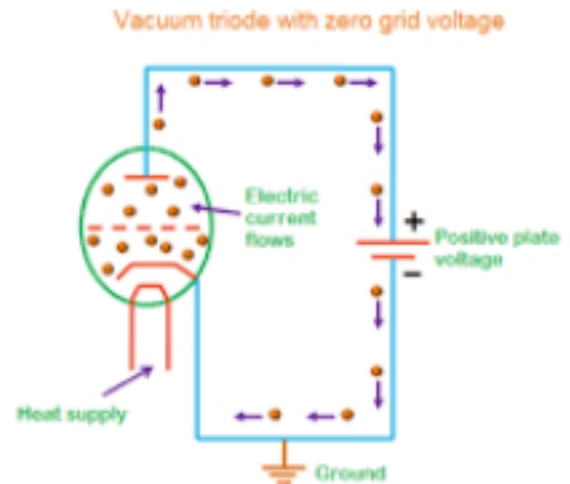


1.4.1 Vacuum Triode with zero grid voltage:

If no voltage is applied to the control grid and positive voltage is applied to the plate, the vacuum triode behaves like a normal vacuum diode, because control grid will not show any effect on the free electrons emitted from the cathode.

If voltage is applied to the control grid, it produces an electric field. In this case, no voltage is applied to the control grid. Hence, the control grid will not produce the electric field to attract or repel the free electrons.

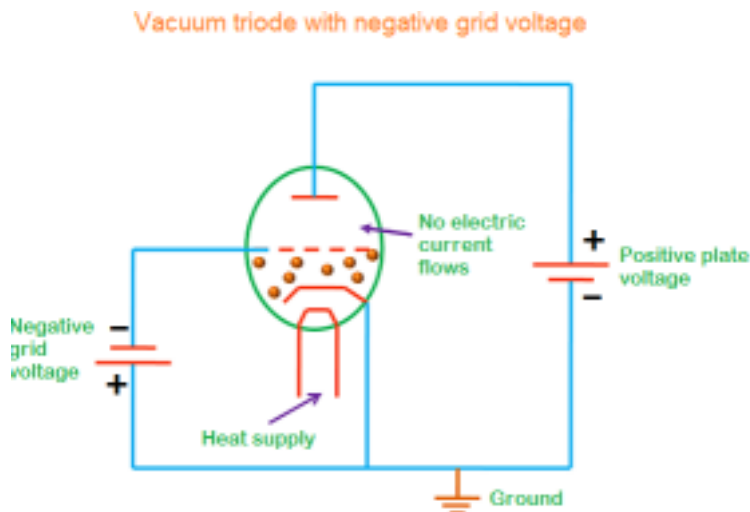
Therefore, the free electrons emitted from the cathode will easily move towards the anode or plate from the openings of the control grid.



Vacuum Triode with Negative grid voltage

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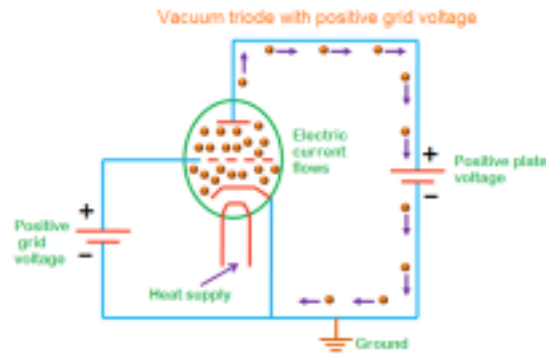
If negative voltage is applied to the control grid without changing the positive plate voltage, no electric current flows in the vacuum triode, because the control grid opposes or repels the free electrons that try to move towards the anode. Because of this supply of negative voltage, the control grid becomes negatively charged. Hence, it produces a negative electric field. On the other hand, free electrons emitted from the cathode are also negatively charged. Hence, free electrons also produce negative electric fields.



We know that, if two like or same charges are placed close to each other they get repelled. Hence, the control grid opposes or repels the free electrons emitted from the cathode. However, a small number of free electrons overcome the negative electric field of the grid and move towards the anode. If the negative voltage applied to the control grid is increased, no electrons will move towards the anode. Hence, no electric current flows in the vacuum triode.

1.4.2 Vacuum Triode with Positive grid voltage:

If positive voltage is applied to the control grid without changing the positive plate voltage, electric current flows in the vacuum triode, because the control grid attracts a large number of free electrons. The free electrons that are attracted towards the control grid will move easily towards the anode.



If positive voltage is applied to the control grid, it becomes positively charged. Hence, it produces a positive electric field towards the free electrons. On the other hand, free electrons emitted from the cathode are negatively charged. Hence, free electrons produce negative electric fields towards the control grid.

We know that, if two opposite charged particles are placed close to each other they get attracted. Hence, the control grid attracts the free electrons. The free electrons that are attracted towards the control grid will easily move towards the anode. The free electrons carry the electric current while moving from cathode to anode.

If the positive voltage applied to the control grid is further increased then even more free electrons are attracted towards the control grid. Therefore, electric current increases with increase in the grid voltage.